blood

721. CLINICAL ALLOGENEIC TRANSPLANTATION: CONDITIONING REGIMENS, ENGRAFTMENT, AND ACUTE TRANSPLANT TOXICITIES: MICROBIODATA, ENDOTHELIAL DAMAGE, AND OPPORTUNISTIC INFECTIONS | NOVEMBER 29, 2018

Pre-Engraftment Gut Colonization with Enterococcus Casseliflavus Improves Survival after Allogeneic Hematopoietic Cell Transplantation

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Abstract

Introduction: Gut bacteria modulate the immune system and influence outcomes of allogeneic hematopoietic cell transplantation (allo-HCT). The presence of *Blautia*, butyrate-producing Clostridia, and *Eubacterium limosum* in the stool has been associated with lower mortality after allo-HCT. We previously reported an unexpected association between pre-HCT gut colonization with intrinsically vancomycin-resistant enterococci (iVRE: *E. gallinarum* and *E. casseliflavus*) and improved overall survival (OS) due to decreased non-relapse mortality (NRM). In an expanded cohort, now also including patients who were iVRE-colonized early post-HCT (before day +14), we demonstrate that the iVRE association with improved OS is specific to *E. casseliflavus*.

Patients and Methods: We studied allo-HCT recipients at our institution who had at least 1 positive rectal swab or stool culture for iVRE between days -14 and +14. New admissions for HCT were screened for gut VRE colonization weekly until discharge. Spectra™ VRE chromogenic agar medium

(ThermoFisher Scientific) was used for species-level identification. Antimicrobial prophylaxis consisted of levofloxacin, acyclovir, and an azole. Cefepime was our empiric antibiotic for neutropenic fever.

Results: 66 (23 with *E. casseliflavus* and 43 with *E. gallinarum*) of the 873 allograft recipients between 2011-2017 met our inclusion criteria. As expected from the constitutive, rather than acquired, vancomycin resistance of iVRE, the groups did not differ in their prior exposure to different classes of antibiotics. There were no significant differences between the groups in patient-, disease-, or transplant-related characteristics, except a higher frequency of sirolimus-based GVHD prophylaxis in the *E. casseliflavus* group (35% vs. 12%, $P = 0.05$). With a median follow up of 30 months, OS was significantly better in patients with *E. casseliflavus* (91% vs. 62% at 3 years, $P = 0.04$), due to lower NRM (0 vs. 18% at 3 years, $P = 0.05$) (**Figures 1A and 1B**). Only 2 patients with *E. casseliflavus* died within 3 years post-HCT, both due to relapse. In contrast, 14 patients with *E. gallinarum* died within 3 years post-HCT: 7 (50%) due to relapse, and 7 (50%) from NRM (4 GVHD, 1 infection, 1 graft failure, and 1 veno-occlusive disease). In multivariable analysis, *E. casseliflavus* gut colonization was significantly associated with reduced all-cause mortality (hazard ratio 0.20, 95% confidence interval 0.04-0.91, $P = 0.04$). There were no significant differences between the groups in 180-day acute grade II-IV GVHD ($P = 0.19$), 1-year chronic GVHD ($P = 0.56$), 100-day bacteremia ($P = 0.59$), or 100-day *Clostridium difficile* infection ($P = 0.79$). To probe the mechanism of this protection against mortality, we mined 15 *E. casseliflavus* and 8 *E. gallinarum* sequenced genomes for 14 shikimate and tryptophan metabolism enzymes. This analysis predicted that *E. casseliflavus* encodes a larger number of enzymes in the tryptophan metabolism pathway (**Figure 1C**). Several compounds in this pathway are ligands for the aryl hydrocarbon receptor (AhR). Signals downstream of AhR augment the gut barrier and modulate the immune system, potentially reducing pathogenic injury and inducing protective immune responses.

Conclusions: Pre-engraftment gut colonization with *E. casseliflavus*, but not *E. gallinarum*, improves survival after allo-HCT. Future mechanistic studies of the interactions between *E. casseliflavus* and the host can guide the development of novel microbiota-oriented therapeutics in this high-risk patient population.

Disclosures

Weisdorf: *Seattle Genetics:* Consultancy; *FATE:* Consultancy; *SL Behring:* Consultancy; *Pharmacyclics:* Consultancy; *Equillium:* Consultancy.

Author notes

* Asterisk with author names denotes non-ASH members.

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